

PH 514: Electronics

Course Code	PH 514
Course Name	Electronics
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	M.Sc. and I-Ph.D. in Physics
Mutual Exclusion	NA
Approval	9th Senate (May 2016)
Revised	10th Senate (Jun 2016) - credits changed; 64th BoA (May 2026) - syllabus revised

Preamble

This course aims to provide students with a comprehensive understanding of the principles of electronic devices, circuits, and instrumentation techniques that are widely used in modern physics laboratories and scientific measurements. It introduces the theoretical foundations of analog, digital, frequency-domain, and mixed-signal electronics, with emphasis on their operating principles, characteristics, and applications in physical science experiments. The course is intended to develop the conceptual framework required to understand electronic systems used in instrumentation and measurement, thereby enabling students to interpret, analyze, and apply electronic concepts in advanced studies and research in physics and related areas.

Course Modules

- **Unit/Topic 1: Amplifiers:** BJT, Classification of Amplifiers, Cascading of amplifiers, Types of power amplifiers, Amplifier characteristics, Feedback in amplifiers, Feedback amplifier topologies, Effects of negative feedback. [10 lectures]
- **Unit/Topic 2: Oscillators and Multivibrators:** Classification and basic principle of an oscillator, Feedback oscillator's concepts, Types of oscillators. [4 lectures]
- **Unit/Topic 3: Field effect transistors:** JFET, FET parameters, MOSFET. [3 lectures]
- **Unit/Topic 4: Operational amplifiers:** OPAMPs, OPAMP applications. [2 lectures]
- **Unit/Topic 5: Boolean algebra and Digital circuit:** Number systems, Boolean algebra, De Morgan's theorem, Logic Gates, Karnaugh Maps [6 lectures]
- **Unit/Topic 6: Combinational circuits:** Adder, Multiplexer, DE multiplexer, Encoder, and Decoder. [5 lectures]
- **Unit/Topic 7: Clock and timing circuit:** Clock waveform, Schmitt Trigger, 555 Timer [2 lectures]
- **Unit/Topic 8: Sequential circuits:** Flip-Flops, Registers, Counters, D/A and A/D conversions [10 lectures]

Laboratory/practical/tutorial Modules: No

Text Books

1. Millman and Halkias, *Integrated electronics*: McGraw-Hill, (2001).
2. A. P. Malvino and D. P. Bates *Electronic Principles*: (7th Edn) McGraw-Hill (2006).

References

1. D. P. Leach, A. P. Malvino and G. Saha, *Digital Principles and Applications*: (6th Edn), Tata McGraw Hill (2007).
2. A. K. Maini, *Digital Electronics-Principles, Devices and Applications*: John Wiley & Sons (2007).